MandiMitra

Explainable mandi price intelligence

Does machine learning beat “tomorrow’s price = today’s price” for wholesale onion and tomato in Jaipur? We tested it with a protocol that cannot pick the winner on the test data.

Vishnu Keer · Vivekananda Global University, Jaipur · Preprint, October 2026

Live app
mandimitra.streamlit.app

Code + data
github.com/vishnu-keer/mandimitra

Historical data. The official series ends on 28 April 2015. Nothing here is a current price or financial, trading or farming advice.

Problem

Mandi prices jump: Jaipur wholesale onion went from Rs. 600–900 per quintal in June 2013 to Rs. 4,500 on 21 October 2013. Price-prediction tools reach farmers and traders, but how they were tested is rarely visible.

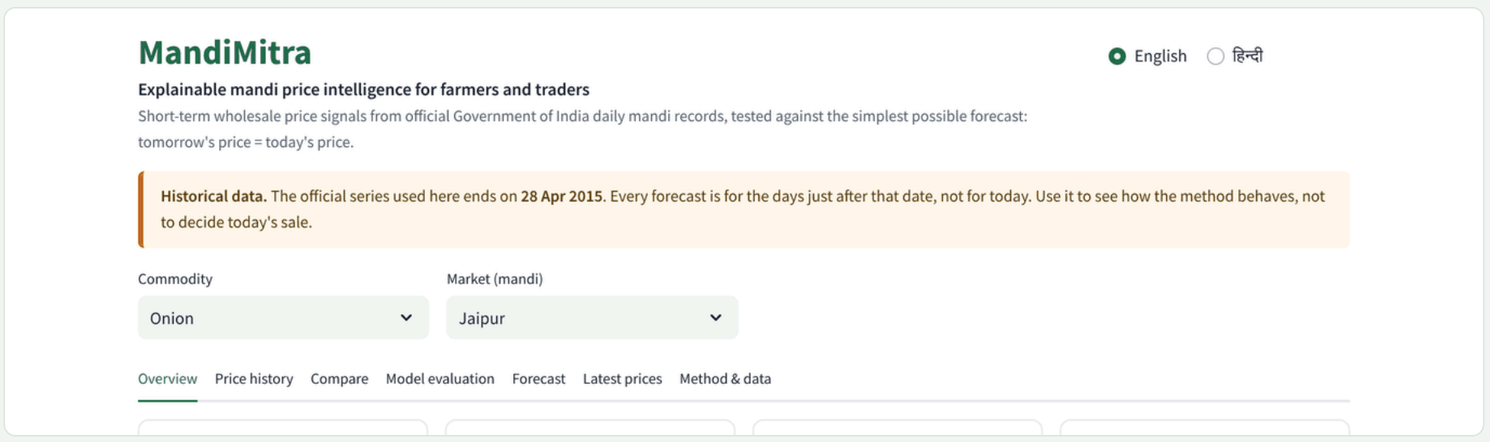
84.8%

of reported Onion/Jaipur days quote exactly the previous price (Tomato 82.8%).

~55%

of calendar days have a reported price at all.

Solution



A bilingual (English / हिन्दी) dashboard: every forecast beside “same as today”, data freshness and stale-quote rates on screen, uncertainty drawn only if it held up, CSV downloads, phone-ready. Free to run.

Architecture and data

Official CSVs	Clean	Features ≤ day t	Choose by CV	Test once	Dashboard
Dept. of Consumer Affairs, data.gov.in; untouched, SHA-256	0 = missing; daily calendar	lags, averages, volatility; leakage-proofed	4 rolling folds, rule fixed in advance	Feb 2014 – Apr 2015, ~200 days per horizon	read-only; trains nothing

Onion and Tomato, Jaipur, 1 Jan 2009 – 28 Apr 2015 (1,002 and 995 usable 1-day rows). Jodhpur and Kota excluded by a 400-row gate set in advance. Licence: GODL-India, with attribution. 154 automated tests; byte-identical rebuilds; CI on every push.

Evaluation

- Candidates: Naive, ARIMA, Random Forest, Gradient Boosting, RF predicting the change.
- Only days the mandi really reported are scored; every model on identical days.
- Embargo: no training target crosses a boundary.
- Diebold–Mariano tests vs Naive, Holm-adjusted.

Verified results — the model chosen in advance never beat “same as today”

Series	Days ahead · chosen by CV	Test RMSE	Naive RMSE	vs Naive
Onion/Jaipur	1 · Naive	94.27	94.27	0.00%
Onion/Jaipur	2 · Naive	110.29	110.29	0.00%
Onion/Jaipur	3 · Naive	127.93	127.93	0.00%
Tomato/Jaipur	1 · RF-delta	127.33	120.51	−5.66%
Tomato/Jaipur	2 · ARIMA	179.10	178.16	−0.53%
Tomato/Jaipur	3 · ARIMA	227.38	226.68	−0.31%

Rs. per quintal. Naive also had the lowest MAE at all six. 80% ranges covered only 55–74% of test outcomes, so no band is drawn.

Significance

0 of 48

Diebold–Mariano tests found any model more accurate than Naive.

p = 0.505

for our first version’s “+7.33%” — which came from picking the winner on the test set.

Limitations

- Historical only (ends April 2015); back-test, not prospective.
- One mandi, two commodities; prices only — no arrivals or weather.
- About 200 test days per horizon: “not distinguishable” is not “equal”.
- No user study yet; no impact is claimed.

Next

- Connect the official AGMARKNET daily price API (client built and tested; needs a free key).
- More markets through the same gate; arrivals and rainfall inputs.
- Score live forecasts prospectively.
- Consented study with 5–10 Jaipur traders and farmers.

Data: Dept. of Consumer Affairs, Govt. of India, via data.gov.in (GODL-India) · Code: MIT · Paper: research/MANDIMITRA_RESEARCH_PAPER.pdf